BLG 454E - Learning from Data
Fall – 2025
-Term Project-

Problem: Emotion Recognition.

Deadline: 01 January 2026, 11:00 PM

The term project has 3 deliverables: The Kaggle competition results, the source code and a report explaining the designed framework and presenting your results. You will be provided with **templates to use** for the report.

Students with the best overall performance and methodology will be invited to collaborate on a research paper based on this project.

1. Kaggle Competition (25 points)

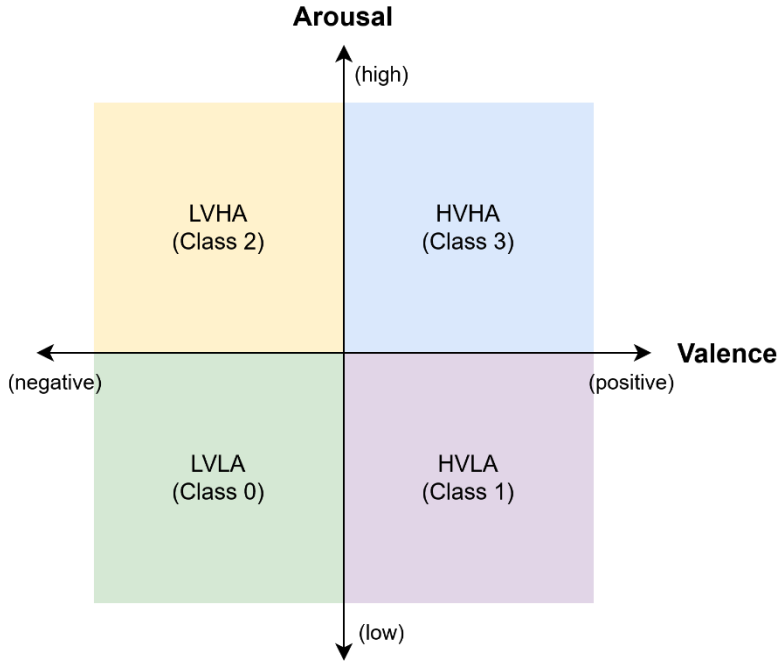
Description

We have created a private class competition on Kaggle. Please click the following link for the term project competition:

https://www.kaggle.com/t/500fa16813f747bc860725061d58100f

Dataset

The dataset you will use in the term project is designed for emotion recognition. For each sample, your goal is to predict which one of four emotion classes (0, 1, 2, or 3) it belongs to. These four classes come from the valence–arousal representation of affect as shown in the figure below. There are two axes, called valence and arousal, and the plane can be split into four regions. These regions are called low valence–low arousal (LVLA), high valence–low arousal (HVLA), low valence–high arousal (LVHA), and high valence–high arousal (HVHA). Each region corresponds to one of the class labels you will predict: class 0 for LVLA**,** class 1 for HVLA**,** class 2 for LVHA**,** and class 3 for HVHA.



In the term project, you will not work with raw dataset directly. Instead, for each sample you are given a 1793-dimensional feature vector extracted from the original dataset. In total you will see 1795 columns: 1793 feature columns, one column (at index −2) that stores the class label (an integer in {0, 1, 2, 3}), and one column (at index −1) that stores the person_id, which tells you from which subject that sample comes.

We will provide a training set and a test set files. The training set contains 22,496 samples and includes features, labels, and person_id. The test set contains 10,656 samples and includes only features.

Goal

There is no restriction on the methods you use. You may apply classical machine learning algorithms, neural networks, feature selection, dimensionality reduction, class rebalancing or model ensembling, and you may perform any preprocessing on the features. Your task is to design a strong classification pipeline that achieves good macro F1-score on the hidden test set, and generalizes to new persons not present in the training set.

Submission Process

To see the performance of your model on test data, submit your predictions of test data to Kaggle in the defined format. Kaggle will calculate and rank the submission scores using the public test data throughout the competition. These scores are publicly visible on public leaderboard. After the competition end, a *private* test data is used to calculate final model performance. Private leaderboard is not released to users until the competition has been closed. Public leaderboard is calculated with 50% of the test data. The final results will be based on the other 50%, so the final standings may be different. Therefore, train your model as general as possible to avoid overfitting on train and public part of the test data.

Scoring Metric

In Kaggle, your submission is evaluated by the F1-Score over the four classes.

Submission File Format

You should submit a csv file with exactly [10656] x [2] entries plus a header row. The file should have exactly 2 columns:

1. ID: [0,…, 10655]
2. Predicted class

```
ID,Predicted
0,3
1,2
2,1
3,0
...
10654,1
10655,0
```

Submission CSVs must have a header row consisting of ID and Predicted as in the sample submission. Using different column names causes a fail in submission process. ID column must include all ID values between [0, 10655].

PS: Your submission will raise an error in cases: you have extra columns (beyond ID and Predicted), extra rows, ID column doesn't consist of integers between [0, 10655], Predicted column includes values other than real-values.

You can download the sample submission file (sampleSubmission.csv) to have a better idea.

Rules

- Every student has to create a Kaggle account and join the competition.
- The project is **individual**. You must work on your own. Team submissions are not allowed.
- Your Kaggle name should make it easy to identify you (Name Surname – Student ID)
- Submission format is explained and a sampleSubmission file (sampleSubmission.csv) is given in the competition webpage.
- You are allowed to use only **Python** programming languages (with jupyter) for the implementation.
- You are only allowed 10 submissions per **DAY**. Start early so you can submit more submissions.
- Academic dishonesty including cheating, plagiarism, and direct copying is unacceptable. Note that your codes and reports will be checked using plagiarism tools!

2. Report (20 points)

Prepare a report in Latex using provided IEEE Conference Paper template. Your report must **not exceed** 3 pages (**one extra page** can be allowed for the **main Figure** illustrating the learning pipeline)!

The report should consist of the following sections:

1. (**2 points) Introduction:** Mention about what and why you did in this project briefly. Give your final score and rank in the competition with your Kaggle name.

2. **(3 points) Datasets:** Explain your methods for data preprocessing in detail.

3. **(10 points) Methods: The how?** Describe each component of your classifier. Include a **main figure** illustrating the key steps of the proposed solution (learning pipeline). Explain how you train and test your model in general. **The why?** Explain **why** you have made selected such components. Give all details about the methods like the algorithms used, parameter tuning, etc.

4. **(4 points) Results and Conclusions: First**, report your **5-fold cross-validation** results. Compare how well your model generalizes when the validation subjects are unseen. Explain your results. **Second**, give your Kaggle score and ranking.

5. **(1 points) References:** The list of references cited in the report. Don't forget the citation to the related reference in the report.

3. Code with 5-fold CV and 5-fold Group K-Fold CV (30 points)

The version of your code that you will upload should implement two different 5-fold cross-validation schemes. The code should take the train data as input, and perform a standard and 5-fold cross-validation and a Group 5-Fold cross-validation, where the groups are defined by the person_id column, for training and validating the designed framework model.

The code will have two outputs: (1) the predicted samples saved in a predictions.csv file (you can use the same Kaggle format to same them), and (2) the average F1 score obtained from 5-fold CV and Group 5-Fold CV.

Important note 1: the code will take in 22496 samples provided to you and perform CV on this set. At this stage, you don't need to use the extra test set that's used in evaluation for Kaggle competition

Tidy up your code as to

- run simply,
- get all necessary inputs as function parameters (train and test data, model parameters),
- produce output, i.e. the submission file (test predictions)
- have explanatory comments

Important note 2: Use the following random anchorization seed:

```
- import random as r
- r.seed(1)
```

Important note 3: For computing the F1 score, once you complete each 5-fold CV scheme, you will end up with predicted classes and ground truth (actual) classes. You can compute the F1 score as follows:

```
- from sklearn.metrics import f1_score as f1s
- f1s(actual,predicted,average='macro')#returns f1 score result for two melted matrices
```

Important note 4: You can use the pandas module to read in your CSV files

(https://pandas.pydata.org/docs/reference/api/pandas.read_csv.html)

```
- import pandas
- pandas.read_csv(filename, header=0, index_col=0)# Read a csv files into DataFrames.
```

4. Project Overall Evaluation

For the project, you will provide a final report in IEEE conference paper format (that is given to you in Latex format). Total score of your project will be calculated as follows:

- The Kaggle competition (55 points)
  - 30 points: 5-fold CV and 5-fold Group K-Fold CV
  - 25 points: Your Kaggle rank
- Report: 20 points
- Code: 25 points
  - Your code should be clean and readable. Implement your code as powerful as possible befitting for a 4th grade student. Weak coding might cause losing 5 to 10 points.

Bonus Marks

Top five students will be rewarded with bonus marks, 20pts, 16pts, 12pts, 8pts and 4pts. respectively, according to the average of the public and private leaderboard scores.

Ninova Submission Policy

- Submit your PDF report, and code in a zip file through Ninova on time.
  - **Unnecessary uploadings (files, pictures, etc.) will be penalized!**
  - **Only put things in your zip file that you are asked to.**
- No late submissions will be accepted.

References

To learn more about Kaggle Competitions, https://www.kaggle.com/docs/competitions

To inspire the decision mechanism for classification, Y. Zhao, X. Cao, J. Lin, D. Yu and X. Cao, "Multimodal Affective States Recognition Based on Multiscale CNNs and Biologically Inspired Decision Fusion Model," in IEEE Transactions on Affective Computing, vol. 14, no. 2, pp. 1391-1403, 1 April-June 2023, doi: 10.1109/TAFFC.2021.3093923.

To inspire subject-independent generalization of a model, M. Ali, F. Al Machot, A. Haj Mosa, M. Jdeed, E. Al Machot and K. Kyamakya, "A Globally Generalized Emotion Recognition System Involving Different Physiological Signals," Sensors, vol. 18, no. 6, p. 1905, 2018, doi: 10.3390/s18061905.

To inspire group-specific / personalized models, M. M. Rahman, M. Trabelsi, H. Uzunalioglu and A. Boyd, "Personalized Mixture of Experts for Multi-Site Medical Image Segmentation," 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Tucson, AZ, USA, 2025, pp. 3172-3184, doi: 10.1109/WACV61041.2025.00314.

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